BAN 100
PROJECT

NAME:

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Problem №1

As an initial step, we excluded the casual variables and *registered* from the regression model. Both variables are components of the outcome variable *count*, and their inclusion would compromise the integrity of the analysis by inducing perfect dependence and artificially inflating model fit. To ensure a valid and interpretable model, we restricted the predictors to external explanatory variables only.

a) Find a multiple regression model for the data.

From the sas output this is our multiple regression line equation

count^=β0+β1season+β2holiday+β3workingday+β4weather+β5temp+β6atemp+β7humidity+β8windspeedcount^=135.79052+22.75882(season)−9.15872(holiday)−1.14953(workingday)+5.93872(weather)+1.84737(temp)+5.63120(atemp)−3.05684(humidity)+0.77762(windspeed)

Number of Observations Read	10886
Number of Observations Used	10886

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	93190870	11648859	479.97	<.0001
Error	10877	263982044	24270		
Corrected Total	10885	357172914			

Root MSE	155.78751	R-Square	0.2609
Dependent Mean	191.57413	Adj R-Sq	0.2604
Coeff Var	81.31970		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	135.79052	8.71016	15.59	<.0001
season	1	22.75882	1.42662	15.95	<.0001
holiday	1	-9.15872	9.27009	-0.99	0.3232
workingday	1	-1.14953	3.31527	-0.35	0.7288
weather	1	5.93872	2.61924	2.27	0.0234
temp	1	1.84737	1.14210	1.62	0.1058
atemp	1	5.63120	1.05057	5.36	<.0001

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
humidity	1	-3.05684	0.09262	-33.00	<.0001
windspeed	1	0.77762	0.19999	3.89	0.0001

b) Interpret the values of the coefficients in the model.

Each regression coefficient represents the expected change in the predicted number of bike rentals associated with a one-unit increase in the corresponding predictor, while holding all

other variables constant. This interpretation follows the standard linear regression framework and allows us to isolate the unique contribution of each variable.

- **Intercept** = 135.79052 Expected bike rentals when all predictors = 0.
- **season** = 22.75882 Moving from one season to the next increases expected rentals by 22.75882.
- **holiday** = –9.15872 Holidays reduce expected rentals by 9.15872
- **workingday** = –1.14953 Working days reduce rentals by 1.14953
- **weathe**r = 5.93872 Worse weather category increases rentals by 5.93872
- **temp** = 1.84737 A one-unit increase in normalized temperature increases rentals by 1.84737
- **atemp** = 5.63120 A one-unit increase in "feels-like" temperature increases rentals by 5.63120
- **humidity** = –3.05684 A one-unit increase in humidity decreases rentals by 3.05684
- **windspeed** = 0.77762 A one-unit increase in windspeed increases rentals by 0.77762.

c) Test whether the model is significant. At the 0.05 level of significance, what is your conclusion?

At the 0.05 level of significance, the regression model as a whole is statistically significant because the ANOVA test gives Pr > F < 0.05. Therefore, we reject the null hypothesis and conclude that at least one predictor is linearly related to the number of bikes rented

Our model summary shows:

$R^2 = 0.2609$ and Adjusted $R^2 = 0.2604$

At the 0.05 significance level:

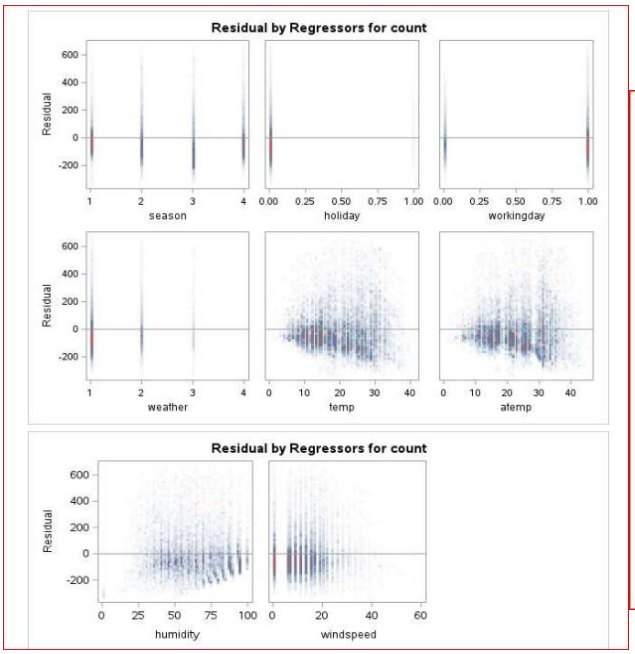
Pr > F < 0.05 → the model is statistically significant

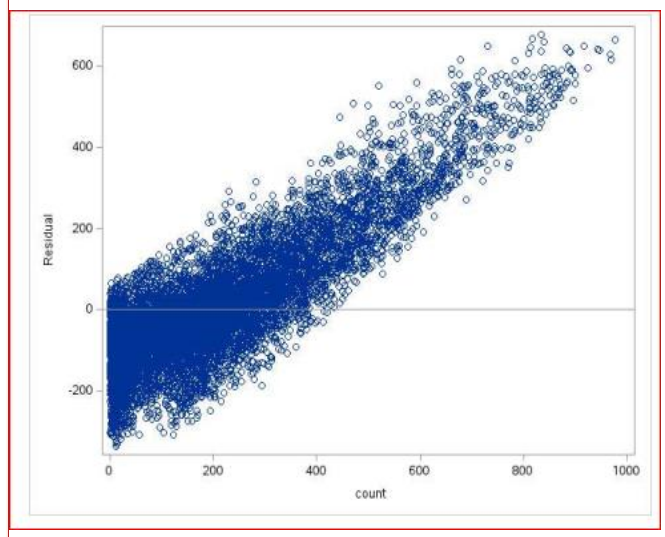
Pr > F ≥ 0.05 → the model is NOT statistically significant

d) Plot the residuals versus the actual values. Do you think that the model does a good job of predicting the number of bikes? Why or why not?

The residual plots reveal substantial vertical dispersion, with values ranging from approximately –200 to +600. This widespread indicates considerable unexplained variability. Moreover, the residuals exhibit a clear pattern of increasing variance as the actual bike count increases, suggesting a violation of the homoscedasticity assumption. Residuals do not cluster tightly around zero and show clear patterns, especially when plotted against the actual count, where residuals increase as the count increases. This indicates unequal variance and systematic underprediction of high bike counts.

Therefore, the model does not do a good job of predicting the number of bikes. The residuals show strong variability and non-random patterns, which is consistent with the low R^2 value (0.2609).





e) Find and interpret the value of R2 for this model.

$R^2 = 0.2609$

Exactly 26.09% of the variation in bike rentals is explained by the predictors. The remaining 73.91% is unexplained. This indicates a weak predictive model

f) Do you think that this model will be useful in helping the planners? Why or why not?

The model is useful for understanding general patterns in bike usage, but it is not reliable for precise forecasting. Planners can use it for insights, not for exact demand prediction.

Even with low predictive accuracy, the model still reveals important trends like:

- rentals increase with season
- rentals increase with atemp
- rentals decrease with humidity
- weather and windspeed matter

g) Test the individual regression coefficients. At the 0.05 level of significance, what are your conclusions?

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	135.79052	8.71016	15.59	<.0001
season	1	22.75882	1.42662	15.95	<.0001
holiday	1	-9.15872	9.27009	-0.99	0.3232
workingday	1	-1.14953	3.31527	-0.35	0.7288
weather	1	5.93872	2.61924	2.27	0.0234
temp	1	1.84737	1.14210	1.62	0.1058
atemp	1	5.63120	1.05057	5.36	<.0001

Program Summary - FinalAss_1.sas

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
humidity	1	-3.05684	0.09262	-33.00	<.0001
windspeed	1	0.77762	0.19999	3.89	0.0001

Significant predictors ($p < 0.05$):

- season ($p < .0001$)
- weather ($p = 0.0234$)
- atemp ($p < .0001$)
- humidity ($p < .0001$)
- windspeed ($p = 0.0001$)

Not significant ($p \geq 0.05$):

- holiday ($p = 0.3232$)
- workingday ($p = 0.7288$)
- temp ($p = 0.1058$)

Conclusion:

Only season, weather, atemp, humidity, and windspeed significantly affect bike rentals. Holiday, working day, and temp do not significantly contribute to the model.

h) If you were going to drop just one variable from the model, which one would you choose? Why? Any other variables you would drop?

We drop variables that are not significant, and not conceptually essential, and possibly correlated with others.

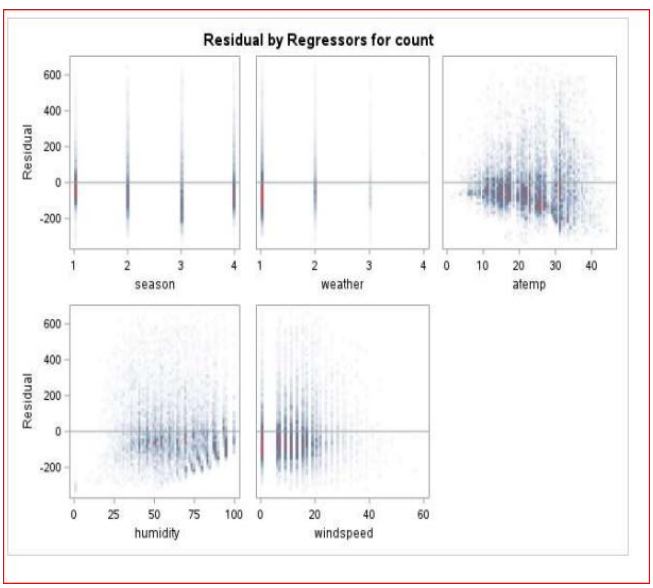
Workingday here will be our best variable to drop because

- P= 0.7288 (highest p-value)
- Almost no effect (–1.14953)
- Conceptually redundant with holiday and season

i) Use stepwise regression to find the best model for the data.

The stepwise regression code uses both forward selection and backward elimination to automatically choose the best subset of predictors. Variables enter the model only if their p-value is below 0.05 (slentry=0.05) and remain in the model only if their p-value stays below 0.05 (slstay=0.05).

Using this method, the final model selected the variables season, weather, atemp, humidity, and windspeed, while holiday, workingday, and temp were removed because they were not statistically significant at the 0.05 level.



Stepwise Selection: Step 7

Variable weather Entered: R-Square = 0.2607 and C(p) = 6.4774

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	93106473	18621295	767.23	<.0001
Error	10880	264066441	24271		
Corrected Total	10885	357172914			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	132.17364	8.25060	6228784	256.64	<.0001
season	22.78971	1.42522	6205836	255.69	<.0001
weather	6.03374	2.61631	129086	5.32	0.0211
atemp	7.30509	0.18389	38301515	1578.09	<.0001
humidity	-3.06681	0.09238	26745970	1101.98	<.0001
windspeed	0.83663	0.19629	440892	18.17	<.0001

Bounds on condition number: 1.4177, 30.135

All variables left in the model are significant at the 0.0500 level.

No other variable met the 0.0500 significance level for entry into the model.

j) Analyze the model you have identified to determine whether it has any problems.

The final stepwise model includes season, weather, atemp, humidity, and windspeed. Although these variables are statistically significant, the model still has several problems.

First, the residual plots show unequal variances and a clear pattern where residuals increase as the actual bike count increases. This indicates that the model violates the constant-variance assumption.

Second, the model has a low R^2 value (0.2609), meaning that it explains only 26.09% of the variation in bike rentals. This suggests that important predictors are missing and that the model has limited predictive accuracy.

Third, multicollinearity is present, particularly between tempe and atemp, although stepwise regression removed temp to reduce this issue, correlation still remains a concern.

Finally, the residual patterns suggest possible nonlinearity, meaning that a simple linear model may not fully capture the true relationships in the data.

Overall, while the model is statistically significant, it has several diagnostic issues and limited predictive power.

k) **Write a memo reporting your findings to your boss. Identify the strengths and weaknesses of the model you have chosen**

MEMO

Subject: Findings from Bike Rental Regression Analysis

Overview

After conducting a multiple regression analysis to identify the key factors influencing daily bike rental counts. After evaluating the full model and applying stepwise regression, the final model retained the following predictors: season, weather, atemp, humidity and windspeed

These variables were statistically significant at the 0.05 level and formed the best-fitting model according to stepwise selection criteria.

<u>Key Findings</u>

1. **Significant Predictors**

- Season and weather conditions have strong effects on bike usage.
- Atemp (feels-like temperature) is a strong positive predictor of demand.
- Humidity has a strong negative effect on rentals.
- Windspeed shows a smaller but statistically significant positive effect.

These variables collectively explain meaningful patterns in bike rental behavior.

2. **Strengths of the Model**

- Statistically Significant Overall: The ANOVA F-test shows the model is significant at the 0.05 level.
- Clear Directional Insights: The model identifies which environmental factors increase or decrease bike usage. This is valuable for planning staffing, maintenance, and resource allocation.
- Simplified Predictor Set: Stepwise regression removed non-significant variables (holiday, workingday, temp), resulting in a cleaner and more interpretable model.

3. **Weaknesses of the Model**

- Low Predictive Power ($R^2 = 0.2609$): The model explains only 26.09% of the variation in bike rentals. This limits its usefulness for precise forecasting.
- Unequal variance: Residual plots show increasing variance as rental counts rise. This violates regression assumptions and reduces reliability of predictions.
- Potential Omitted Variables: Important factors such as time of day, day of week, special events, and station-level effects are not included. Their absence likely contributes to the low R^2.
- Possible Nonlinearity: Residual patterns suggest that relationships may not be purely linear.

Conclusion

The final regression model is statistically valid and provides useful insights into how weather and seasonal conditions influence bike rental demand. However, due to its low explanatory power and diagnostic issues, it should be used for understanding trends, not for accurate forecasting.

To improve predictive performance, we recommend exploring additional variables and potentially testing more flexible modeling approaches

Problem №2

a) **Write the logistic regression equation relating to Age and Survived**

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-0.0567	0.1736	0.1068	0.7438
Age	1	-0.0110	0.00533	4.2310	0.0397

The logistic regression model relating Age to Survived: Survived = −0.0567 − 0.0110(Age)

b) Estimate the probability of surviving the passenger with the average Age 30.

The probability of surviving the passenger with the average Age 30.

Obs	PassengerId	Survived	Class	Name	Sex	Age	SiblingSpouse	ParentChild	Ticket	Fare	Cabin	Embarked	_LEVEL_	prob_survived
892	.	.	.			30	.	.		.			1	0.40477

The probability is calculated using:
p = e^(−0.0567 − 0.0110 × 30) / (1 + e^ (−0.0567 − 0.0110 × 30))
p = 0.40477
As a result, the probability of survival for a passenger aged 30 is approximately 0.4048 (40.48%).

c) Suppose we want to check who has a 0.50 or higher probability of surviving. What is the average age to achieve this level of probability?

When p = 0.50:
log (0.50 / (1 − 0.50)) = 0
0 = −0.0567 − 0.0110(Age)
Age = −(−0.0567) / −0.0110
Age = −5.174
As a result, the calculated age is −5.174, which is not realistic.

Interpretation:
This implies that there is no realistic age where the chance of survival reaches 0.50. Since the coefficient for Age is negative, the survival probability decreases as age increases.

The average probability to achieve 0.4 or higher probability of surviving

Obs	PassengerId	Survived	Class	Name	Sex	Age	SiblingSpouse	ParentChild	Ticket	Fare	Cabin	Embarked	_LEVEL_	prob_survived
1	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25		S	1	0.42607
2	3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925		S	1	0.41538
3	8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075		S	1	0.48035
4	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333		S	1	0.41272
5	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708		C	1	0.44764
6	11	1	3	Sandstrom, Miss. Marguerite Rut	female	4	1	1	PP 9549	16.7	G6	S	1	0.47488
7	13	0	3	Saundercock, Mr. William Henry	male	20	0	0	A/5. 2151	8.05		S	1	0.43144
8	15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14	0	0	350406	7.8542		S	1	0.44764
9	17	0	3	Rice, Master. Eugene	male	2	4	1	382652	29.125		Q	1	0.48035
10	19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele)	female	31	1	0	345763	18		S	1	0.40213

To verify the results, we adjusted the probability to 0.4 and determined that the maximum predicted probability of survival in this model is 0.48035 (at Age = 0).

Consequently, in this specific model (using only the Age variable), no passenger achieves a predicted survival probability of 0.5 or higher.

d) What is the estimated odds ratio? What is the interpretation?

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Age	0.989	0.979	0.999

Odds Ratio = e^ (−0.0110) = 0.98910

The estimated odds ratio for Age is 0.989. This means that for every one-year increase in age, the odds of surviving the Titanic decrease by approximately 1.1% (since 1−0.989=0.011), holding all other factors constant. In other words, older passengers had lower odds of survival compared to younger ones.

e) For the Titanic data, estimate a comprehensive logistic regression model with equations with other variables included.

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-0.9461	0.2751	11.8271	0.0006
Sex	female	1	2.6277	0.2148	149.6888	<.0001
Age		1	-0.0448	0.00822	29.6145	<.0001
Class	1	1	2.6526	0.2858	86.1240	<.0001
Class	2	1	1.2382	0.2453	25.4723	<.0001
SiblingSpouse		1	-0.3802	0.1215	9.7888	0.0018

The following variables were included in the model: sex, age, class, siblingspous.

The logistic regression model relating these variables to Survived:

$$Survived = -0.9461 + 2.6277(Sex_female) - 0.0448(Age) + 2.6526(Class_1) + 1.2382(Class_2) - 0.3802(SiblingSpouse)$$

f) Interpret the model and provide your overall conclusions

1. Gender (Sex: Female): 2.6277

Interpretation: This is one of the strongest predictors in the model. The positive coefficient indicates that being female significantly increased the odds of survival compared to being male.

Context: This aligns with the historical "women and children first" protocol during the evacuation.

2. Passenger Class (Class 1&2): 2.6526 (Class 1) and 1.2382 (Class 2)

Interpretation: Both coefficients are positive, meaning passengers in higher classes were more likely to survive than those in Class 3 (the reference group).

Key Insight: The survival advantage for Class 1 was more than double that of Class 2 relative to the baseline, highlighting a strong socio-economic bias in survival rates.

3. Age: -0.0448

Interpretation: The negative coefficient indicates an inverse relationship between age and survival. For every additional year of age, the log-odds of survival decreased.

Context: Younger passengers (children) were prioritized, while older passengers faced lower chances of rescue.

4. SiblingSpouse: -0.3802

Interpretation: This variable has a negative impact. As the number of siblings or spouses on board increased, the individual's probability of survival decreased.

Context: This may suggest that passengers traveling with family members were delayed by searching for or staying with their relatives during the chaos.

Overall Conclusions:

- Socio-Demographic Dominance: Survival was not random. It was heavily dictated by a combination of gender and social class. A female passenger in first class had the highest statistical probability of survival.
- Impact of Class: The disparity between classes is stark. Even when controlling for age and gender, being in the third class was a significant disadvantage, likely due to the location of the cabins and limited access to the boat deck.
- Model Efficiency: By considering only significant variables, we have achieved a parsimonious model. This means the model is simple yet highly effective, focusing only on the "critical drivers" of survival.

- Final Verdict: The model successfully confirms historical accounts of the disaster: "Privilege" (Class) and "Policy" (Women/Children First) were the primary determinants of who lived and who died.

g) Please provide the syntax and output to answer each question

Problem №3: Capital punishment

a. Why the odds ratios are different? Explain it

From the two logistic regression models ran with different coding schemes for race:

Model 1: White = 0, Black = 1 and Model 2: Black = 0, White = 1

The odds ratios differ because changing the coding reverses the direction of the comparison.

In Model 1, the coefficient compares Black defendants to White defendants.

Odds ratio = 0.34

Black defendants have 0.34 times the odds of receiving capital punishment compared to White defendants.

In Model 2, the coding is reversed, so the comparison is White defendants to Black defendants.

Odds ratio = 2.95

White defendants have 2.95 times the odds of receiving capital punishment compared to Black defendants.

b. Show the relation between the odd ratios and coefficient.

The odds ratio is simply the exponential of the logistic regression coefficient. Changing the sign of the coefficient flips the odds ratio to its reciprocal.

For Model 1:

- Coefficient = –1.081
- Odds ratio = $e^{-1.081}$=0.34

For Model 2:

- Coefficient = +1.081
- Odds ratio = e1.081=2.95

c. What are the differences between model 1 and model 2. What is your key conclusion?

Differences

- The coding of race is reversed.
- The sign of the coefficient changes (–1.081 vs +1.081).
- The odds ratio changes from 0.34 to 2.95.
- The interpretation direction changes (Black vs White vs White vs Black).

Key Conclusion

Both models tell the exact same story: White defendants have higher odds of receiving capital punishment than Black defendants. The difference in odds ratios is only due to how the race variable was coded.